

GIG CASHFLOW INDEX

Comprehensive Architectural Blueprint

ML System | Backend | Frontend Dashboard

Version 1.0 — April 2026

Mission Financial inclusion for 300M+ gig workers	Method Temporal volatility decomposition + fraud detection	Output GCI Score 300–900 + explainability report	Alignment RBI Digital Lending + Account Aggregator
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Section 1 — ML Models & Training

1.1 Synthetic Data Generation

For the MVP, all training and testing data is synthetically generated using AI. The data must faithfully simulate the financial behaviour of Indian gig workers across diverse occupations, city tiers, and seasonal patterns.

1.1.1 Borrower Profile Characteristics

Characteristic	Specification	Rationale
Occupations	Swiggy/Zomato rider, Ola/Rapido driver, freelancer, daily-wage tailor, street vendor, plumber	Covers full gig spectrum — platform and informal
City tiers	Tier 1 (Mumbai, Bengaluru), Tier 2 (Pune, Jaipur), Tier 3 (Nashik, Madurai)	Income levels and seasonal patterns differ by tier
Tenure range	6 months to 6 years active gig history	Tests model across thin-file and thick-file borrowers
Income range	Rs 8,000 – Rs 45,000 per month	Covers MUDRA loan target demographic fully
Age distribution	22–55 years, skewed 25–38	Reflects actual Indian gig worker demographics
Borrower profiles	500 total — 400 clean, 80 suspicious, 20 high-risk	Matches estimated 5–10% manipulation rate in population

1.1.2 Transaction Data Schema

Each synthetic transaction record contains the following fields:

Field	Type	Description
transaction_id	UUID	Unique identifier for each transaction
borrower_id	UUID FK	Links to borrower profile table
timestamp	TIMESTAMP	Full datetime — enables hourly/daily/weekly analysis
amount	DECIMAL(10,2)	Transaction amount in INR
direction	ENUM(credit, debit)	Money in or money out

Field	Type	Description
counterparty_id	VARCHAR(64)	Hashed UPI ID or bank account reference
counterparty_type	ENUM	employer / merchant / peer / unknown / platform
payment_mode	ENUM	UPI / NEFT / IMPS / cash_deposit / platform_payout
category	VARCHAR(32)	rent / grocery / fuel / salary / tip / transfer / loan
description	VARCHAR(128)	Free-text narration (used for category inference)
balance_after_txn	DECIMAL(10,2)	Running balance — critical for liquidity scoring
is_recurring	BOOLEAN	System-flagged: appears monthly with stable amount
festival_window	VARCHAR(32)	Null or: diwali / ipl / navratri / holi / monsoon

1.1.3 Data Volume Requirements

Dataset	Borrowers	Avg Txns/Borrower	Total Records	Purpose
Training set	350	420	~147,000	Train all 4 models
Validation set	75	420	~31,500	Hyperparameter tuning
Test set	75	420	~31,500	Final evaluation, no leakage
Adversarial set	50 (fraud only)	420	~21,000	Stress-test manipulation detection
Total	500+	420 avg	~231,000+	Full pipeline

Minimum 12 months of history per borrower required for seasonal decomposition to be statistically valid.
Fraud profiles injected at 16% rate in training data (higher than real-world to ensure model learns minority class).
Each fraud profile contains exactly one of the 5 manipulation patterns to enable isolated detection testing.

1.2 Model Architecture — Four-Model Pipeline

GCI uses four specialized models in a sequential pipeline. Each model solves one well-defined sub-problem. This modularity allows independent retraining, evaluation, and replacement.

Model	Type	Algorithm	Output	Replaces in Post-MVP
M1 — Temporal Decomposer	Time-series	STL Decomposition + Fourier	Explained vs residual variance ratio	Prophet / SARIMA
M2 — Anomaly Detector	Unsupervised ML	Isolation Forest	Anomaly score 0.0–1.0 + label	Autoencoder (LSTM)
M3 — Fraud Rule Engine	Hard statistical rules	Z-score + Counterparty graph	Flag bitmask (5 flags)	Graph Neural Network
M4 — PD Scorer	Supervised ML	Gradient Boosting (XGBoost)	Probability of Default 0.0–1.0	LightGBM + SHAP ensemble

M1 — Temporal Decomposer (Core Innovation)

This is the novel contribution of GCI. STL (Seasonal-Trend decomposition using LOESS) decomposes each borrower's monthly income time series into three components:

- Trend component — long-term earning trajectory (growing, stable, declining)
- Seasonal component — recurring calendar-driven cycles (festivals, cricket, monsoon)
- Residual component — true unexplained variance — the only component penalised in scoring

India Festival Calendar Layer: Diwali (Oct/Nov), Navratri (Sep/Oct), Holi (Mar), IPL season (Mar-May), Monsoon (Jun-Sep), Harvest months (Oct-Nov, Feb-Mar).

Cohort ceiling applied per occupation x city tier: spike beyond 99th percentile of cohort for that festival = unexplained excess.

Cross-year consistency: current festival spike compared to same festival in prior 2 years. Anomaly if $>1.5\times$ prior average.

M2 — Isolation Forest Anomaly Detector

Unsupervised model trained on clean borrower feature vectors. Anomalous profiles are isolated in fewer random partitions — this gives an anomaly score.

- Contamination parameter: 0.08 (8%) — tunable as real data accumulates
- Feature space: 12 engineered cashflow features (see Section 1.4)
- Output: anomaly_score (0.0–1.0) mapped to Clean / Suspicious / High Risk
- Threshold: score > 0.65 = Suspicious, score > 0.80 = High Risk

M3 — Statistical Fraud Rule Engine

Five hard rules operating independently of ML. Binary triggers. Fast, explainable, and directly RBI-compliant:

Flag ID	Rule Name	Detection Logic	Threshold
F1	Income spiking	Income in last 90 days vs own 12-month mean. Cohort-bounded.	> 99th percentile of occupation cohort for that window
F2	Expense suppression	Count of recurring expense categories in assessment window vs baseline.	3+ recurring categories disappear from prior stable baseline
F3	Round-trip transactions	Same amount credit+debit to same counterparty_id within 72 hours.	>= 2 instances in 60-day window
F4	Synthetic regularity	Std deviation of weekly inflows: last 90 days vs 24-month baseline.	Ratio < 0.10 (10x more regular than historical)
F5	Dormant activation	Transaction frequency: last 90 days vs prior 18-month average.	Frequency delta > 8x baseline activity

M4 — Probability of Default (XGBoost)

Supervised binary classifier trained on labelled synthetic borrower outcomes. Takes CVD-adjusted features as input.

- Algorithm: XGBoostClassifier with class_weight balanced
- Train/test split: 70/15/15 (train/val/test) with stratified sampling on default label
- Cross-validation: 5-fold stratified CV
- Primary metric: AUC-ROC (target > 0.82 on test set)
- Secondary metrics: Precision, Recall, F1 at 0.5 threshold
- Saved as: models/pd_model.pkl via joblib

1.3 GCI Score Formula

GCI = base_score
+ (income_stability_score × 0.30)
+ (liquidity_buffer_score × 0.25)
+ (repayment_capacity_score × 0.20)
+ (expense_discipline_score × 0.15)
- (residual_volatility_penalty × 0.10)
× manipulation_penalty_multiplier

manipulation_penalty_multiplier: Clean=1.0 | Suspicious=0.75 | High Risk=0.50

Score range: 300 (high risk) to 900 (excellent). Bands: 300–499 Poor | 500–619 Fair | 620–719 Good | 720–799 Very Good | 800–900 Excellent

1.4 Feature Engineering — RBI & Research-Backed Parameters

All features are derived from raw transaction data. Feature selection is grounded in RBI's Digital Lending Guidelines (2022), the Account Aggregator framework, and academic literature on cashflow-based credit risk.

Temporal Analysis Features (M1 input + M4 input)

Feature	Formula / Derivation	Regulatory Basis
avg_monthly_income	Mean of monthly credit totals over 12 months	RBI AA Framework — income verification
income_volatility_raw	Std dev / mean of monthly income (CV)	Research: Guvenen et al. 2021 earnings volatility
cvd_residual_variance	Residual component from STL decomposition	Novel — GCI contribution (CTVS framework)
explained_variance_ratio	1 - (residual_var / total_var)	Novel — measures calendar explainability
income_trend_slope	Linear regression slope over 12-month income series	CGAP 2023: earning trajectory signals creditworthiness
seasonal_amplitude	Max - min of seasonal component (STL)	Captures festival sensitivity of occupation
weekly_income_variance	Std dev of weekly inflow totals over 52 weeks	RBI: repayment capacity assessment guideline
peak_to_trough_ratio	Max monthly income / min monthly income	Standard cashflow risk metric
festival_uplift_ratio	Festival month income / cohort 90th percentile ceiling	>1.0 triggers excess unexplained flag
cross_year_festival_consistency	Current festival spike / mean of prior 2 year same festival	Novel — detects camouflaged fraud in seasonal windows

Liquidity & Repayment Features (M4 input)

Feature	Formula / Derivation	Regulatory Basis
liquidity_buffer_weeks	Avg end-of-month balance / avg weekly expenditure	RBI: minimum 2-week buffer recommendation
expense_income_ratio	Total monthly debit / total monthly credit (avg)	Standard debt-service ratio proxy
recurring_expense_count	Count of stable monthly debit signatures (amount $\pm$ 10%)	Signals financial responsibility and stability
emi_coverage_ratio	Avg monthly surplus / proposed EMI amount	Direct repayment capacity test
overdraft_frequency	Count of months where balance_after_txn < 500	Distress signal — CFPB research 2023
savings_rate	(Income - Expenses) / Income averaged over 12 months	Standard financial health metric
platform_income_share	Platform payouts / total income (for gig workers)	Legitimacy signal — platform income is verifiable

Anomaly Detection Features (M2 + M3 input)

Feature	Formula / Derivation	Fraud Pattern Detected
mom_income_change_rate	Month-over-month % change in income, rolling 3M	Income spiking (F1)
new_counterparty_ratio_spike	New counterparty count in spike window / baseline count	Dark money source injection
round_trip_pair_count	Count of same-amount in/out to same counterparty_id within 72h	Round-trip washing (F3)
expense_signature_delta	Recurring expense categories present in baseline minus assessment window	Expense suppression (F2)
weekly_inflow_std_ratio	Std dev last 90 days / std dev 24-month baseline	Synthetic regularity (F4)
dormancy_score	Transaction frequency last 90 days / frequency prior 18 months	Dormant account activation (F5)
inflow_source_diversity	Shannon entropy of counterparty_type distribution	Detects unnatural uniformity or explosion
festival_excess_ratio	Income during festival / cohort 99th percentile ceiling	Seasonal camouflage fraud

1.5 Miscellaneous ML Additions

Cohort Management System

Every borrower is assigned to a micro-cohort for relative scoring. Cohort definition:

- Occupation category (platform delivery / informal services / freelance / trading)
- City tier (Tier 1 / Tier 2 / Tier 3)
- Income band (< Rs 12k / Rs 12–25k / Rs 25–40k / > Rs 40k)
- Tenure band (< 1 year / 1–3 years / 3+ years)

Cohort percentile distributions are pre-computed and stored in PostgreSQL. Every anomaly threshold is cohort-relative, not population-absolute.

Model Versioning & Retraining

- All models versioned with MLflow or simple timestamp tagging in /models/ directory
- Retraining triggered when: new data volume exceeds 20% of training set, or F1 score drops > 5% on rolling validation
- A/B testing framework: new model serves 10% of requests until statistically validated

Explainability Layer (RBI Compliance)

- SHAP (SHapley Additive Explanations) values computed for every PD model prediction
- Top 5 contributing features extracted per borrower and stored in gci_scores table
- Human-readable explanation template: 'Score driven primarily by [feature] contributing +X points'
- Required by RBI Digital Lending Guidelines (2022) Section 7.3: lenders must explain algorithmic credit decisions

Section 2 — Backend Architecture

2.1 Technology Stack

Layer	Technology	Version	Justification
API Framework	FastAPI	0.111+	Async Python, auto OpenAPI docs, pydantic validation, native ML model serving
Language	Python	3.11+	Single language across ML + backend — no context switching, shared utilities
Database	PostgreSQL	16+	ACID compliance, JSONB for flexible score metadata, excellent Python ORM support
ORM	SQLAlchemy	2.0+	Async ORM, Alembic migrations, type-safe queries
ML Runtime	Scikit-learn + Statsmodels	1.4+ / 0.14+	Isolation Forest, STL decomposition — production-stable
Boosting	XGBoost	2.0+	PD model training and inference
Data Processing	Pandas + NumPy	2.0+ / 1.26+	Feature engineering pipeline
Caching	Redis	7+	Cache GCI scores (TTL 30 days), rate limiting, session tokens
Auth	JWT (python-jose)	3.3+	Stateless auth for borrower and lender portals
Task Queue	Celery + Redis	5.3+	Async model inference for large batch uploads
Containerisation	Docker + Docker Compose	Latest	Reproducible deployment, all services in single compose file
Cloud (MVP)	AWS EC2 t3.medium	—	Cost-effective MVP. Scale to ECS Fargate post-validation
Monitoring	Prometheus + Grafana	—	API latency, model drift, error rates

2.2 System Architecture

Request Flow:

Borrower/Lender Portal (React) --> FastAPI Gateway --> Auth Middleware (JWT)
--> Route Handler --> Celery Worker (async) --> ML Pipeline
--> [M1 Temporal Decomposer] --> [M2 Isolation Forest] --> [M3 Rule Engine] --> [M4 XGBoost PD]
--> GCI Score Calculator --> SHAP Explainer --> PostgreSQL (persist)
--> Redis (cache) --> API Response (JSON) --> React Dashboard

2.3 API Endpoints

Method	Endpoint	Description	Auth
POST	/api/v1/upload-transactions	Upload CSV or JSON transaction file for a borrower. Triggers async pipeline.	Borrower JWT
POST	/api/v1/borrower/register	Register new borrower profile with KYC metadata.	Open + OTP
GET	/api/v1/borrower/{id}	Fetch borrower profile and summary stats.	Borrower or Lender JWT
GET	/api/v1/gci/{borrower_id}	Fetch full GCI score report with SHAP explanations.	Lender JWT
GET	/api/v1/temporal/{borrower_id}	Fetch temporal decomposition data (trend, seasonal, residual arrays).	Lender JWT
GET	/api/v1/anomaly/{borrower_id}	Fetch anomaly score, flag breakdown, and manipulation risk label.	Lender JWT
GET	/api/v1/stress/{borrower_id}	Fetch stress simulation results (3 scenarios).	Lender JWT
GET	/api/v1/model/feature-importance	Return global SHAP feature importance for current PD model.	Admin JWT
POST	/api/v1/lender/batch-score	Batch score multiple borrowers. Returns paginated results.	Lender JWT
GET	/api/v1/health	Health check for all services (DB, Redis, model availability).	Open

2.4 Database Schema

Table: borrowers

Column	Type	Notes
id	UUID PK	Primary key
full_name	VARCHAR(128)	Encrypted at rest
phone_hash	VARCHAR(64)	SHA-256 of phone — used for dedup, no PII stored
occupation	VARCHAR(64)	Enum: delivery / driver / freelance / informal / trading
city	VARCHAR(64)	
city_tier	SMALLINT	1 / 2 / 3
income_band	VARCHAR(16)	low / mid / high
cohort_id	VARCHAR(64)	Derived: occupation_tier_band
consent_timestamp	TIMESTAMP	AA framework consent record
created_at	TIMESTAMP	

Table: transactions

Column	Type	Notes
id	UUID PK	
borrower_id	UUID FK	→ borrowers.id, CASCADE DELETE
timestamp	TIMESTAMP	Indexed for time-series queries
amount	DECIMAL(10,2)	
direction	VARCHAR(6)	credit / debit
counterparty_id	VARCHAR(64)	Hashed — no raw UPI IDs stored
counterparty_type	VARCHAR(32)	employer / merchant / peer / unknown / platform
payment_mode	VARCHAR(16)	UPI / NEFT / IMPS / cash / payout
category	VARCHAR(32)	ML-inferred from description
balance_after_txn	DECIMAL(10,2)	
festival_window	VARCHAR(32)	Nullable — populated by festival calendar service
is_recurring	BOOLEAN	System-flagged

Table: gci_scores

Column	Type	Notes
id	UUID PK	
borrower_id	UUID FK	→ borrowers.id
gci_score	SMALLINT	300–900
pd_score	DECIMAL(5,4)	0.0000–1.0000
income_stability_score	DECIMAL(5,2)	0–100
liquidity_buffer_score	DECIMAL(5,2)	0–100
expense_discipline_score	DECIMAL(5,2)	0–100
repayment_capacity_score	DECIMAL(5,2)	0–100
manipulation_label	VARCHAR(16)	clean / suspicious / high_risk
manipulation_penalty	DECIMAL(4,2)	1.0 / 0.75 / 0.50
anomaly_score	DECIMAL(5,4)	Isolation Forest output
flag_bitmask	SMALLINT	Binary: bit 0=F1, bit 1=F2, ... bit 4=F5
shap_top5	JSONB	Array of {feature, value, shap_value} for top 5 drivers
explained_variance_ratio	DECIMAL(5,4)	CVD output — key research metric
generated_at	TIMESTAMP	
valid_until	TIMESTAMP	generated_at + 30 days
model_version	VARCHAR(16)	Traceability

2.5 Model-to-Backend Connection

All models are loaded once at application startup into memory using a singleton ModelRegistry pattern. No disk reads on each request.

Model loading (startup): ModelRegistry.load() reads all .pkl files from /models/ directory into a dict.	
Inference (per request): Celery worker calls ModelRegistry.get('pd_model').predict(features).	
Feature pipeline:	FeatureEngineer class in ml/features/ transforms raw transactions to feature vector.
Async execution:	Upload endpoint returns task_id immediately. Client polls GET /task/{id} for result.

Model updates:
downtime.

New model file dropped into /models/ triggers hot-reload without

2.6 Security

Concern	Implementation
PII protection	Phone numbers hashed (SHA-256). UPI counterparty IDs hashed. No raw identifiers stored.
Data at rest	PostgreSQL column-level encryption for name, account references. AES-256.
Data in transit	HTTPS enforced. TLS 1.3. HSTS header on all responses.
Authentication	JWT tokens. Access tokens expire in 1 hour. Refresh tokens in 7 days. Stored in httpOnly cookies.
Rate limiting	Redis-backed rate limiter: 100 requests/minute per lender, 10 uploads/day per borrower.
Input validation	Pydantic models validate all request bodies. SQL injection impossible via SQLAlchemy ORM.
AA Compliance	Consent timestamp stored per borrower. Data deletion endpoint available (GDPR-equivalent).
Secrets management	All secrets via environment variables. Never in code. Docker secrets in production.

2.7 Scalability

- Horizontal scaling: FastAPI is stateless — scale behind NGINX load balancer or AWS ALB
- Database scaling: Read replicas for lender dashboard queries. Write-primary for transaction ingestion.
- Celery workers: Scale worker count independently from API servers based on queue depth
- Redis cluster: For production, upgrade to Redis Cluster for high-availability caching
- Model serving: Post-MVP, migrate to dedicated TorchServe or BentoML for sub-10ms inference at scale

Section 3 — Frontend Dashboard

3.1 Technology Stack

Technology	Purpose	Justification
React 18 + Vite	Core UI framework + build tool	Fast HMR, modern JSX, industry standard for dashboards
TypeScript	Type safety across components	Catches prop errors at compile time — critical for financial data display
Recharts	Primary charting library	React-native, composable, excellent line/bar/area charts for time-series
D3.js	Advanced custom charts	Heatmaps, box plots, custom temporal decomposition visualization
Tailwind CSS	Utility-first styling	Rapid, consistent, professional — no CSS file management
shadcn/ui	Component library	Pre-built accessible components — tables, cards, badges, dialogs
TanStack Query	Server state management	Auto-caching, background refetch, loading states for API calls
React Router v6	Tab and page routing	Clean URL-based navigation between dashboard tabs
Axios	HTTP client	Interceptors for JWT injection, unified error handling
Framer Motion	Animations	Score reveals, chart transitions — professional fintech feel

3.2 Tab 1 — Anomaly & Fraud Detection Dashboard

Primary audience: Risk officers, NBFC compliance teams, internal analysts.

Layout & Components

Component	Chart Type	Data Source	What it shows
Manipulation Risk Banner	Status badge + score gauge	/api/v1/anomaly/{id}	Clean / Suspicious / High Risk label, penalty multiplier, Isolation Forest score

Component	Chart Type	Data Source	What it shows
Flag Status Panel	5-row checklist with indicators	/api/v1/anomaly/{id}	Each of F1–F5 flags: triggered or clean, with brief reason text
Anomaly Score Gauge	Radial gauge chart (Recharts RadialBar)	/api/v1/anomaly/{id}	0.0–1.0 Isolation Forest score with colour bands (green/amber/red)
Income Source Heatmap	D3 calendar heatmap (months × weeks)	/api/v1/temporal/{id}	Transaction volume by week — reveals dormant activation pattern visually
Counterparty Network	D3 force-directed graph	/api/v1/anomaly/{id}	Counterparty relationships — visual round-trip detection
Round-trip Transaction Log	Filterable data table (shadcn/ui)	/api/v1/anomaly/{id}	Each detected circular transaction pair with amount, dates, counterparty
Expense Signature Timeline	Stacked area chart (Recharts)	/api/v1/temporal/{id}	Monthly recurring vs non-recurring expense — suppression visible as area shrinkage
Festival Window Overlay	Annotated line chart (Recharts)	/api/v1/temporal/{id}	Income line with festival period shading — camouflage fraud visible as excess above band

Colour Coding Standard (Tab 1)

- Green (#15803D background): Clean flag — no anomaly detected
- Amber (#B45309 background): Suspicious — Isolation Forest score 0.65–0.80 or one hard rule
- Red (#B91C1C background): High Risk — score > 0.80 or two+ hard rules triggered
- Blue shading on charts: Calendar-explained windows (festival, IPL, monsoon)
- Red shading on charts: Unexplained anomalous windows

3.3 Tab 2 — Temporal Analysis & Lender Report

Primary audience: Lenders, NBFCs, MUDRA loan officers. Language and framing must match financial industry standards.

Layout & Components

Component	Chart Type	Data Source	What it shows
GCI Score Hero Card	Large score display + band label	/api/v1/gci/{id}	Score 300–900, band (Poor/Fair/Good/Very Good/Excellent), valid until date
Score Breakdown Radar	Radar / Spider chart (Recharts)	/api/v1/gci/{id}	5 dimensions: Income Stability, Liquidity, Repayment Capacity, Expense Discipline, Tenure
Temporal Decomposition Chart	3-panel stacked line chart (Recharts)	/api/v1/temporal/{id}	Panel 1: Raw income. Panel 2: Seasonal component. Panel 3: Residual (risk signal only)
Variance Attribution Pie	Donut chart (Recharts PieChart)	/api/v1/temporal/{id}	Festival % Weekly cycle % Residual % — the core CVD visualization
12-Month Income Trend	Area chart with confidence band (Recharts)	/api/v1/temporal/{id}	Trend line + upper/lower confidence interval. Festival windows shaded.
Stress Test Results	Grouped bar chart (Recharts BarChart)	/api/v1/stress/{id}	3 scenarios side by side: Normal / 25% drop / 40% drop — repayment probability each
Liquidity Buffer Timeline	Line chart with threshold line	/api/v1/temporal/{id}	End-of-month balance / weekly expense ratio. RBI 2-week buffer threshold marked.
Feature Importance (SHAP)	Horizontal bar chart (Recharts)	/api/v1/gci/{id}	Top 5 positive and negative drivers of GCI score in plain English labels
Lender Recommendation Box	Card with structured text	/api/v1/gci/{id}	Suggested loan amount, EMI, tenure, seasonal notes, review triggers
Cohort Comparison	Box plot (D3)	/api/v1/gci/{id}	Borrower score vs cohort distribution — percentile rank shown

Financial Language Standards (Tab 2)

All labels and descriptions must use industry-standard terminology as used in NBFC and bank credit appraisal memos:

Internal ML Term (do NOT show) avg_monthly_income	Lender-facing Label (SHOW this) Average Net Monthly Income (ANMI)
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income_volatility_raw cvd_residual_variance liquidity_buffer_weeks pd_score anomaly_score	Income Volatility Index (IVI) Adjusted Risk Variance (ARV) Liquidity Cover Ratio (LCR) Probability of Default (PD) — % Manipulation Risk Score
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3.4 Dashboard Specifications

Responsive Layout

- Desktop (> 1280px): Two-column grid. Charts full-width. Score card pinned right.
- Tablet (768–1280px): Single column. Charts stack vertically.
- Mobile: Read-only summary view. Full report PDF export button.

Performance Requirements

- Initial load: < 2 seconds (Vite code splitting, lazy-loaded chart libraries)
- API response displayed: < 500ms after fetch completes (no layout shift)
- Chart render: < 100ms for up to 500 data points (Recharts handles this natively)
- D3 heatmap: Virtualized for > 365 days of data

Export & Sharing

- PDF export: React-PDF or html2canvas snapshot of Tab 2 for lender records
- CSV export: Raw temporal data for lender's own analysis
- Shareable link: JWT-protected report URL with 7-day expiry

Accessibility & Compliance

- WCAG 2.1 AA — all charts have text alternatives and ARIA labels
- Colour-blind safe palette — all status indicators use shape + colour (not colour alone)
- RBI disclosure: 'This report is a supplementary risk tool. Final decision rests with the lender.' — shown on every report

Section 4 — Complete Pipeline Summary

End-to-end data flow:
1. Borrower consents (AA framework) + uploads bank statement / UPI export
2. FastAPI validates file → Celery queues async processing task
3. Data Cleaner: null removal, timestamp normalization, amount standardization, category inference
4. Festival Calendar Service: tags each transaction with festival_window if applicable
5. Feature Engineer: computes all 25 features from cleaned + tagged transactions
6. M1 — STL Temporal Decomposer: splits income series into trend + seasonal + residual
7. M2 — Isolation Forest: scores feature vector for anomaly probability
8. M3 — Rule Engine: evaluates F1–F5 hard flags independently
9. Combined fraud verdict: manipulation_label + penalty_multiplier assigned
10. M4 — XGBoost PD Model: predicts probability of default on CVD-adjusted features
11. GCI Score Calculator: applies formula with all scores + penalty multiplier
12. SHAP Explainer: extracts top 5 feature contributions for explainability report
13. Stress Simulator: Monte Carlo 500 simulations × 3 income scenarios
14. Results persisted to PostgreSQL (gci_scores table)
15. Redis cache updated with 30-day TTL
16. FastAPI returns complete GCI report JSON to React dashboard
17. React renders Tab 1 (anomaly) + Tab 2 (temporal/lender report) with all charts

Section 5 — Research Paper & Patent Claim

Novel Contribution: Causal Volatility Decomposition (CVD) for Gig Worker Credit Scoring
Claim: We demonstrate that 55–85% of income volatility in Indian gig worker transaction streams
is causally attributable to verifiable calendar events (festivals, seasons, demand cycles), and
therefore constitutes spurious risk signal under conventional credit models.
We propose CVD — a seasonal decomposition framework that:
(a) separates context-explained variance from true stochastic residuals,
(b) applies cohort-bounded magnitude ceilings to detect seasonal camouflage fraud,
(c) uses cross-year festival consistency to catch single-year manipulation.
We show CVD-adjusted GCI outperforms raw cashflow baselines by recovering creditworthiness
for gig workers systematically underscored by volatility penalisation.
Target: ACM FAccT 2026 AAAI AI for Social Good Workshop SSRN FinTech working paper

— End of Architectural Blueprint —